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Contents

2 Coupled Oscillations **2**

2.1 Review and Examples of Oscillations with 1 Degree of Freedom (a single mass system) . . . 2

2.1.1 Summary of elementary concepts and definitions in oscillations . . 2

2.1.2 Example . . . 5

2.2 Mass-Spring Systems – Longitudinal and Transverse Oscillations, Slinky and Small Angle Approximations . . . 7

2.2.1 Longitudinal oscillations in a mass-two-spring system . . . 7

2.2.2 Transverse oscillations in a mass-two-spring system . . . 9

2.3 Free Oscillations of Systems with Two Degree of Freedom . . . 13

2.3.1 Introduction . . . 13

2.3.2 Example: Two-dimensional harmonic oscillator . . . 15

2.3.3 Systematic method of solution for coupled oscillations . . . 16

2.4 Oscillations of two coupled masses . . . 19

2.4.1 Longitudinal oscillations of two coupled masses . . . 19

2.4.2 Transverse oscillations of two coupled masses . . . 22

2.4.3 Coupled Pendulums . . . 24

2.5 Beats . . . 27

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Chapter 2

Coupled Oscillations

2.1 Review and Examples of Oscillations with 1 Degree of Freedom (a single mass system)

Contents

2.1 Review and Examples of Oscillations with 1 Degree of Freedom (a single mass system) 2
2.2 Mass-Spring Systems – Longitudinal and Transverse Oscillations, Slinky and Small Angle Approximations 7
2.3 Free Oscillations of Systems with Two Degree of Freedom 13
2.4 Oscillations of two coupled masses 19
2.5 Beats 27

2.1.1 Summary of elementary concepts and definitions in oscillations

- Here, we discuss many examples of the motion of a system of particles or masses that has been given an initial external disturbance (we call it an *excitation*) and is allowed to oscillate freely without further influence.
- Such oscillations are called *free or natural oscillations*.
- The oscillations in the systems are described by some physical quantity whose displacement from its equilibrium value varies with position in the system and time.
  - ◇ Mechanical examples involve moving parts which may be regarded as point masses subject to *restoring forces*. The physical quantity is the *displacement* of the mass at the point x, y, z from its *equilibrium position*.
  - ◇ *Equilibrium position* is the point at which restoring forces on the masses are *zero*.
  - ◇ Displacements are denoted by various symbols in various book; they vary based on the context.
    - ⋆ E.g: $x(t), y(t)$ for x and y positions as functions of time, $I(t)$ for current oscillations in an LC-circuit, $E_x(z,t)$ for x-component of electric field of electromagnetic oscillations propagating along z-axis that vary in position and time, and so on.

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- ⋆ For oscillations in 1D (*one-dimension*) we will denote the displacement of a single point mass by

$$\psi(t).$$

- ⋆ The oscillations of a continuous string or an elastic rod or electric or magnetic fields along some direction may be denoted by

$$\psi(x,y,z,t).$$

Sometimes it is also called a *wave function*.

- ⋆ Oscillations in 2D or 3D is denoted by the vector

$$\vec{\psi}(x,y,z,t).$$

For electromagnetic waves electric and magnetic fields are described by vector fields $\vec{E}(x,y,z,t)$ and $\vec{B}(x,y,z,t)$.

- *Free Oscillations* of Systems with One Degree of Freedom – Nomenclature
  - ◇ Examples:
    - ⋆ A pendulum oscillating in a plane can be described by an angle ψ the string makes with the vertical.
    - ⋆ a mass on a spring described by the extension $x(t)$ of the mass from the equilibrium position.
    - ⋆ In an LC circuit, current oscillations are described by a single time-varying quantity $I(t)$.
      Their configuration at any time is specified by a single quantity. They are said to have *one degree of freedom*.
  - ◇ **Degrees of Freedom** (**DoF**):
    - ⋆ If a system has one moving part or one mass, and is its motion is described by a single variable, say, ψ, we say, the system has *one degree of freedom*.
    - ⋆ If the system has two moving parts or two masses that oscillate, say, (x,y) or (ψ_a,ψ_b), we say the system has *two degrees of freedom (DoF)*.
    - ⋆ If the systems has n moving parts or masses, the system is said to have n DoF.
  - ◇ *Harmonic oscillation*. For all systems with one degree of freedom, the displacement or the quantity of interest from its equilibrium value has the simple same time dependence:

$$\psi(t) = A\cos(\omega t + \varphi). \tag{2.1}$$

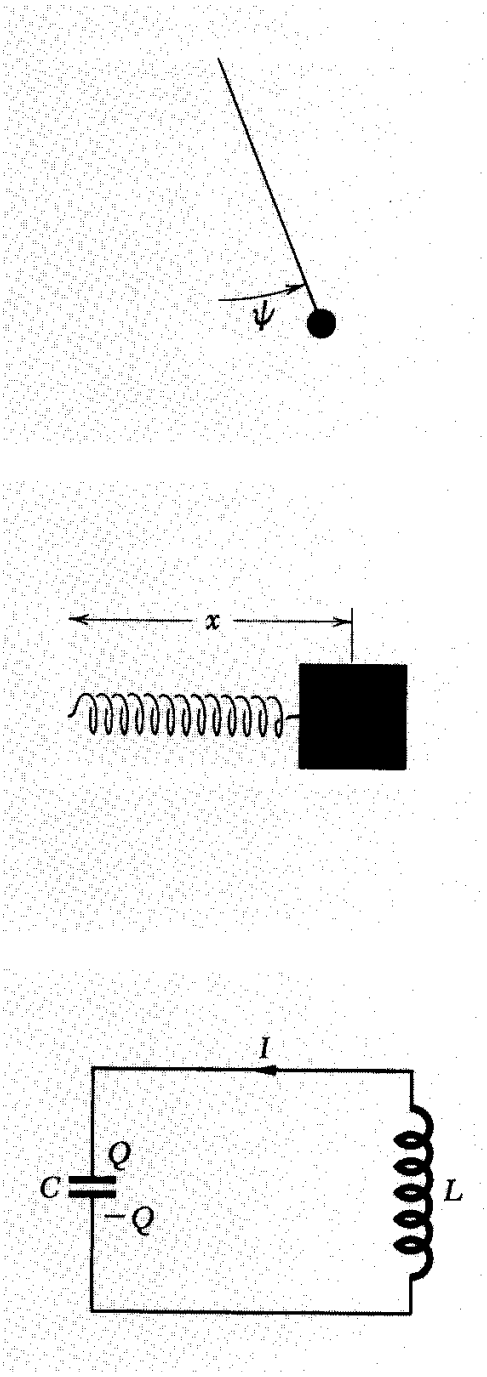


Fig. 2.1: Simple oscillating systems of 1 degree of freedom.

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Such a motion is called a harmonic oscillation. Here, $\psi(t)$ may represent (as illustrated in the figure on the right)

- ⋆ angular displacement of a pendulum bob, or
- ⋆ linear displacement of a mass attached to a spring, or
- ⋆ current or charge in a capacitor in LC-oscillations.

◇ ***Nomenclature***:

- ⋆ In Eq.(2.1): A is a positive constant called the *amplitude*.
- ⋆ ω is the *angular frequency*, measured in radians/sec.
- ⋆ *frequency*, measured in cycles/sec or hertz (abbreviated as cps, or Hz):

$$\nu = \omega/2\pi. \tag{2.2}$$

- ⋆ The inverse of ν is called the *period* T, which is given in seconds per cycle:

$$T = \frac{1}{\nu}. \tag{2.3}$$

- ⋆ The phase constant φ corresponds to the choice of the zero of time or an appropriate initial condition. In many cases we can always "reset the clock" so that $\varphi = 0$ and then we may write

$$\psi = A\cos\omega t \qquad \text{or} \qquad \psi = A\sin\omega t.$$

- ***Restoring force*** F

◇ In systems such as that of a spring-mass system, when the mass is pulled away from the resting or equilibrium position, the tendency of the stretched spring to contract and pull the mass back to the equilibrium position is called the restoring force.

- ⋆ In a pendulum motion, the return force is gravity
- ⋆ For LC circuit oscillations, return force is due to repulsion between electrons.

◇ In the mathematical language, when ψ increases, the restoring force tries bring ψ back to zero (0) by imparting suitable velocity $d\psi/dt$ to the moving part.

◇ Greater the ψ is, the stronger the return force.

- **Physical meaning of** ω^2. The angular frequency of oscillation is

$$\boxed{\omega^2 = \text{restoring force per unit displacement per unit mass.}} \tag{2.4}$$

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2.1.2 Example

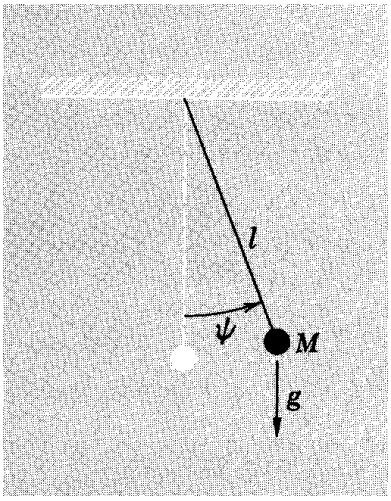


Fig. 2.2: Simple pendulum

Example 2.1. Simple Pendulum

Assumptions: For the pendulum illustrated in the figure on the right, we assume that

- bob of mass M is considered as point mass
- string or rod of length is massless
- length l is a constant.
- pendulum swings in a fixed vertical plane from a rigid support
- motion is friction-less and free (undamped)

Kinematics

- displacement ψ is the angle (in radians) from the vertical.
- arc length of the circular path traced by bob: $l\psi$.
- the instantaneous tangential velocity: $l\dfrac{\mathrm{d}\psi}{\mathrm{d}t} = l\dot{\psi}$.
- tangential acceleration: $l\dfrac{\mathrm{d}^2\psi}{\mathrm{d}t^2} = l\ddot{\psi}$

Dynamics and Newtons law and Equation of Motion

- tangential component of the restoring force: $-Mg\sin\psi$.
- Application of Newton's 2nd law gives the equation of motion (**EoM**)

$$Ml\frac{\mathrm{d}^2\psi}{\mathrm{d}t^2} = -Mg\sin\psi. \tag{2.5}$$

Small angle or small oscillation approximation

- For small oscillations, the the sine function can be linearized for simplicity of mathematics. For this we use Taylor expansion

$$\sin\psi = \psi - \frac{1}{3!}\psi^3 + \frac{1}{5!}\psi^5 - \cdots. \tag{2.6}$$

- When ψ is sufficiently small, in certain situations, say, $\psi = 0.1\,\mathrm{rad}$, $\sin\psi = 0.0998 \approx 0.1 = \psi$. In such cases, we ignore all terms in Eq.(2.6) except the first one, ψ:

$$\sin\psi \approx \psi, \qquad \text{when } |\psi| \ll \pi/2.$$

- The EoM then becomes

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}t^2} = -\frac{g}{l}\psi. \tag{2.7}$$

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- The general solution is the harmonic oscillation is

$$\psi(t) = A\cos(\omega t + \varphi). \tag{2.8}$$

- The corresponding angular velocity and angular acceleration are

$$\dot{\psi}(t) = \frac{\mathrm{d}\psi(t)}{\mathrm{d}t} = -\omega A\sin(\omega t+\varphi), \tag{2.9}$$

$$\ddot{\psi}(t) = \frac{\mathrm{d}^2\psi(t)}{\mathrm{d}t^2} = -\omega^2 A\cos(\omega t+\varphi) = -\omega^2\psi \tag{2.10}$$

- Substituting angular acceleration in Eq. (2.10) into the EoM in Eq.(2.7), we get

$$-\omega^2\psi = -\frac{g}{l}\psi \qquad \Rightarrow \qquad \omega^2 = \frac{g}{l}.$$

We can now write the EoM in the form

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}t^2} = -\omega^2\psi. \tag{2.11}$$

- The square of the angular frequency has the same meaning as defined above:

$$\begin{aligned}\omega^2 &= \text{return force per unit displacement per unit mass}\\ &= \frac{Mg\psi}{(l\psi)M} = \frac{g}{l}.\end{aligned}$$

- Amplitude and phase in terms of initial displacement and velocity: The initial angular displacement and initial angular velocity are

$$\begin{aligned}\psi(0) = \psi_0 &= A\cos\varphi\\ \dot{\psi}(0) = \dot{\psi}_0 &= -\omega A\sin\varphi.\end{aligned}$$

From this we get,

$$\begin{aligned}A &= \left[\psi_0^2 + \left(\dot{\psi}_0/\omega\right)^2\right]^{1/2},\\ \cos\varphi &= \frac{\psi_0}{A}\\ \sin\varphi &= -\frac{\dot{\psi}_0}{\omega A}\\ \varphi &= \tan^{-1}\left(-\frac{\dot{\psi}_0}{\omega\psi_0}\right),\end{aligned}$$

where the angle φ is such that $(x,y) = (\cos\varphi, \sin\varphi)$ is a point P on a unit circle. The value of φ must be chosen carefully either in the range $[0, 2\pi)$ or $(-\pi, \pi]$.

Exercise 2.1. Assume $g = 9.8\,\mathrm{m/s^2}$. Take $l = 1.2\,\mathrm{m}$, $\psi(0) = 0.05\,\mathrm{rad}$, $\dot{\psi}(0) = 0.02\,\mathrm{rad/s}$.

The following algorithm will help determine the correct value of φ using a calculator.

- Case (i), P is in the 1st quadrant, i.e, both $x, y > 0$, the value returned by the calculator is correct:

$$\varphi = \text{calculator value of } \tan^{-1}\left[-\dot{\psi}/(\omega\psi_0)\right].$$

- Cases (ii) and (iii), P is in the 2nd or 3rd quadrant, i.e, $x < 0$, regardless of the sign of y:

$$\begin{aligned}\varphi &= \text{calculator value of } \tan^{-1}\left[-\dot{\psi}/(\omega\psi_0)\right]\\ &\quad +\pi \text{ (rad) or } 180^\circ.\end{aligned}$$

- Case (iv), P is the 4th quadrant, i.e, both $x > 0$ and $y < 0$:

$$\begin{aligned}\varphi &= \text{calculator value of } \tan^{-1}\left[-\dot{\psi}/(\omega\psi_0)\right]\\ &\quad +2\pi \text{ (rad) or } 360^\circ.\end{aligned}$$

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1. Find ω, with appropriate quadrant, ν, T, A.
2. Calculate $\cos\varphi$ and $\sin\varphi$ and mark the location of the point$(\cos\varphi, \sin\varphi)$ on the xy-plane. Hence find φ correctly.
3. Plot $\psi(t)$ for $0 \le t \le 2T$.

2.2 Mass-Spring Systems – Longitudinal and Transverse Oscillations, Slinky and Small Angle Approximations

2.2.1 Longitudinal oscillations in a mass-two-spring system

Example 2.2. Mass and two springs – longitudinal oscillations

Please refer the illustration and configuration in the figure on the right.

- A mass M slides without friction with springs of spring constant K connected on its both sides and connected to the wall.
- Lengths:
  - ◇ natural lengths of the springs: a_0
  - ◇ equilibrium length of the springs: $a > a_0$
  - ◇ position of the mass z
  - ◇ displacement from equilibrium will be called : $\psi = z - a$.
  - ◇ length of the left spring: $l_1 = z = a + \psi$
  - ◇ change in the length of the left spring: $\Delta l_1 = z - a_0 = a + \psi - a_0$
  - ◇ length of the right spring: $l_2 = 2a - z = a - \psi$
  - ◇ change in the length of the right spring: $\Delta l_2 = a - \psi - a_0$.
- Position, velocity, acceleration

$$z = a + \psi, \qquad \dot{z} = \dot{\psi}, \qquad \ddot{z} = \ddot{\psi}.$$

- Forces:
  - ◇ equilibrium tension in each spring in equilibrium:

$$T_0 = K(a - a_0).$$

  - ◇ Tension in the spring 1 on the left when M is displaced:

$$T_1 = K(a + \psi - a_0).$$

  - ◇ Restoring force from the spring 1:

$$F_1 = -T_1 = -K(a + \psi - a_0).$$

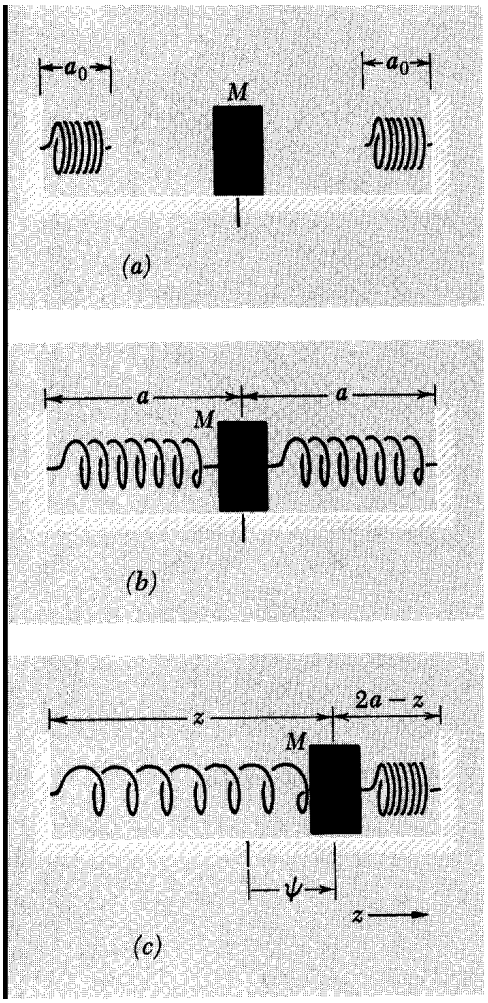


Fig. 2.3: Longitudinal oscillations.
(a) Springs relaxed and unattached.
(b) Springs attached, at equilibrium position.
(c) General configuration.

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$F_1 < 0$ means that the restoring force is in the $-z$ direction. This happens when $T_1 > 0$, i.e., with the spring (attached the left end) is extended.

- ◇ Tension in the spring 2 on the right when M is displaced:

$$T_2 = K(a - \psi - a_0).$$

- ◇ Restoring force from spring 2:

$$F_2 = T_2 = K(a - \psi - a_0).$$

- ◇ Net force on M:

$$\begin{aligned} F_z &= F_1 + F_2 = -T_1 + T_2 \\ &= T_2 - T_1 \\ &= K[a - \psi - a_0] - K(a + \psi - a_0) \\ &= -2K\psi. \end{aligned}$$

- Newton's law and EoM:

$$M\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = M\frac{\mathrm{d}^2 \psi}{\mathrm{d}t^2} = F_z = -2K\psi. \tag{2.12}$$

This can be written in the std form

$$\frac{\mathrm{d}^2 \psi}{\mathrm{d}t^2} = -\omega^2 \psi, \tag{2.13}$$

with

$$\omega^2 = \frac{2K}{M} \tag{2.14}$$

- Note that the frequency is independent of the extent of equilibrium extension of the springs.
- The solution to equation of motion (2.13):

$$\psi(t) = A\cos(\omega t + \phi).$$

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2.2.2 Transverse oscillations in a mass-two-spring system

Example 2.3. Mass and springs-transverse oscillations – Transverse Oscillations

A mass M is attached to rigid support via two massless springs of spring constant K each. We consider the transverse (perpendicular to line connecting the support points) oscillations. Its geometry and other parameters are illustrated in the figure above.

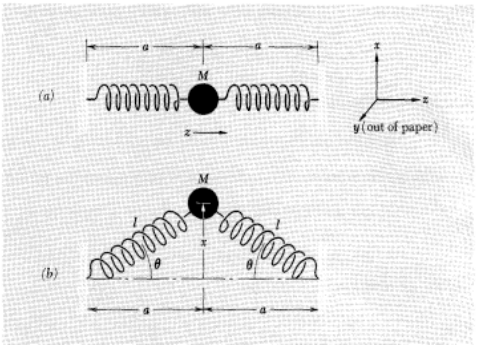


Fig. 2.4: Transverse oscillations. (a) Equilibrium configuration. (b) General configuration (for motion along x).

Lengths, Displacements, Spring Tensions, Forces, Equation of Motion:

- Initial position of the mass: along the z axis at $z = a$.
- natural length of springs: a_0
- equilibrium length of springs in the horizontal position: a
- extension in the spring in the horizontal equilibrium position:

$$\Delta l = a - a_0.$$

- equilibrium tension in the springs:

$$T_0 = K(a - a_0) \tag{2.15}$$

- displacement: x, along the vertical direction (see the coordinate system in Fig.2.4).
- stretched length of each spring: $l = (a^2 + x^2)^{1/2}$
- angle the spring makes with the z axis: θ
- relations among angles and lengths:

$$\cos\theta = \frac{a}{l}, \qquad \sin\theta = \frac{x}{l}, \qquad \tan\theta = \frac{x}{a}.$$

- tension in each spring when M is displaced as shown:

$$T = K(l - a_0) \tag{2.16}$$

- vertical component of the sum of the restoring forces from *two* springs:

$$F_x = -2T\sin\theta = -2K(l - a_0)\frac{x}{l} = -2Kx\left(1 - \frac{a_0}{l}\right). \tag{2.17}$$

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- Applying the Newton's II law gives the EoM:

$$\begin{aligned} M\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= F_x = -2T\sin\theta = -2K(a-a_0)\frac{x}{l} \\ &= -2Kx\left(1-\frac{a_0}{l}\right) \qquad (2.18) \\ &= -2Kx\left(1-\frac{a_0}{\sqrt{a^2+x^2}}\right). \qquad (2.19) \end{aligned}$$

- This is a non-linear equation and cannot be solved. Two kinds of approximations are possible.

Approximation Type 1 – Slinky approximation

- When the natural length of the spring is much less than the stretched length of the spring:

$$a_0 \ll l.$$

- Then we can ignore the 2nd term a_0/l in the parenthesis in the RHS of EoM in (2.18). The EOM then becomes

$$M\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -2Kx\underbrace{\left(1-\frac{a_0}{l}\right)}_{\approx 1} \approx -2Kx \quad \Rightarrow$$

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x,$$

with

$$\omega^2 = \frac{2K}{M}.$$

- Since, for $a_0 = 0$, the equilibrium tension simplifies to

$$T_0 = K(a-a_0) = Ka,$$

and we can express ω^2 in terms of T_0 simply as

$$\omega^2 = \frac{2K}{M} = \frac{2T_0}{Ma}.$$

- The general solution: $x = A\cos(\omega t+\phi)$.

Approximation Type 2 – Small Oscillation approximation

- When we cannot ignore a_0, but the vertical displacement x is much less than the equilibrium length a of the spring,

$$|x| \ll a,$$

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we can make small oscillation or angle approximation.

- We can also arrive at this result as follows. From $F_x = -2T\sin\theta$, for small x,

$$\frac{x}{l} = \sin\theta \approx \theta \approx \tan\theta = \frac{x}{a}. \tag{2.20}$$

This amounts to simply setting $l \approx a$. More details are given at the end of this section.

- The EoM can now be brought to the std form

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= -2\frac{K}{M}\left(1-\frac{a_0}{l}\right)x \\ &\approx -2\frac{K}{M}\left(1-\frac{a_0}{a}\right)x \\ &= -2\frac{K(a-a_0)}{Ma}x \\ &= -\frac{2T_0}{Ma}x, \\ &= -\omega^2 x,\end{aligned}$$

where, as defined earlier,

$$T_0 = K(a-a_0)$$

is the tension in the spring in equilibrium position. Therefore,

$$\omega^2 = \frac{2K}{M}\left(1-\frac{a_0}{a}\right) = \frac{2T_0}{Ma} \equiv \omega_{\mathrm{tr}}^2.$$

- Thus, the frequency of the transverse oscillations are

$$\boxed{\omega_{\mathrm{tr}}^2 = \frac{2K}{M}\left(1-a_0/a\right) = \frac{2T_0}{Ma}, \qquad T_0 = K(a-a_0).}$$

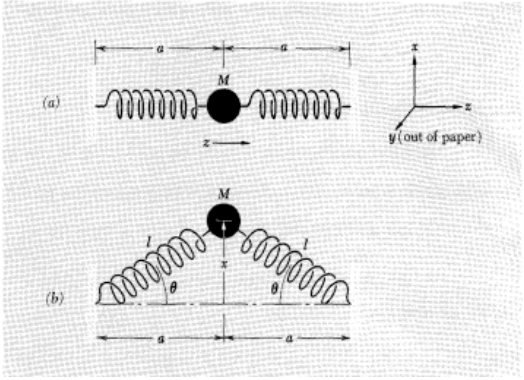
Comparison of angular frequencies of the longitudinal and transverse modes.

- For longitudinal oscillations,

$$\omega_{\mathrm{long}}^2 = \frac{2K}{M}.$$

  - ◇ The ratio of frequencies for longitudinal to transverse oscillations is

$$\frac{\omega_{\mathrm{long}}}{\omega_{\mathrm{tr}}} = \frac{1}{\left(1-a_0/a\right)^{1/2}} > 1.$$



- **Exercise 2.2.** Consider the spring-mass system shown the figure on the right. The mass M can move over a smooth either along the

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z axis along the line connecting the support points or in the perpendicular x direction. Assume the springs are massless. Given parameters are $a_0 = 5\,\text{cm}$, $a = 50\,\text{cm}$, $K = 0.2\,\text{N/m}$, $M = 50\,\text{g}$.

(a) Find ω_{long} (the angular frequency of longitudinal oscillations). Are any approximations involved here?

(b) Find and ω_{tr} for transverse oscillations in the cases (i) slinky approximation (ii) small oscillation approximation.

(c) Find the ratio $\omega_{\text{long}}/\omega_{\text{tr}}$ – use the small angle approximation.

Details of the Small Oscillation Approximation

- The non-linear term in $1/l$ can be simplified as follows. First note that

$$\frac{a}{l} = \frac{a}{(a^2+x^2)^{1/2}} = \left[1 + \left(\frac{x}{a}\right)^2\right]^{-1/2}.$$

If we let $\epsilon = (x/a)^2$, then, using the Taylor expansion $(1+\epsilon)^{-1/2} = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \cdots$, gives us

$$x\left(1 - \frac{a_0}{l}\right) = x\left\{1 - \frac{a_0}{a}\left[1 - \frac{1}{2}\frac{x^2}{a^2} + \cdots\right]\right\} = \left(1 - \frac{a_0}{a}\right)x + \mathcal{O}(x^3).$$

- The linear approximation here means dropping all non-linear terms on the RHS:

$$x\left(1 - \frac{a_0}{l}\right) \approx x\left\{1 - \frac{a_0}{a}\right\},$$

which amounts to simply approximating $l \approx a$, as mentioned above.

- **Special Case: Free length equals the equilibrium length,** $a = a_0$. In this special case, $T_0 = K(a - a_0) = 0$ and hence the linear term vanishes. The EoM, up to the lowest non-zero term, has a restoring force that is cubic in x:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{Ka_0}{Ma^3}x^3.$$

This is clearly non-linear and cannot be solved easily. Hence, for a linear approximation, we will require that $a_0 < a$.

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2.3 Free Oscillations of Systems with Two Degree of Freedom

2.3.1 Introduction

- Some fascinating examples of systems having two or more degrees of freedom.
  - ◇ double pendulum, coupled pendulums, coupled spring-mass systems with two more masses
  - ◇ coupled LC circuits
  - ◇ molecules (H_2,ammonia, etc), elementary particles, such as K mesons, neutrinos.
- If the system takes two variables for its description, say, ψ_a and ψ_b, then we say the system has two degrees of freedom.
  - ◇ For double and coupled pendulums, they denote angles with the verticals of two rods
  - ◇ For spring-mass systems, they denote longitudinal or transverse displacements from equilibrium positions of the masses
  - ◇ For LC circuits, with two inductors and two or more capacitors, they denote currents through the inductors.

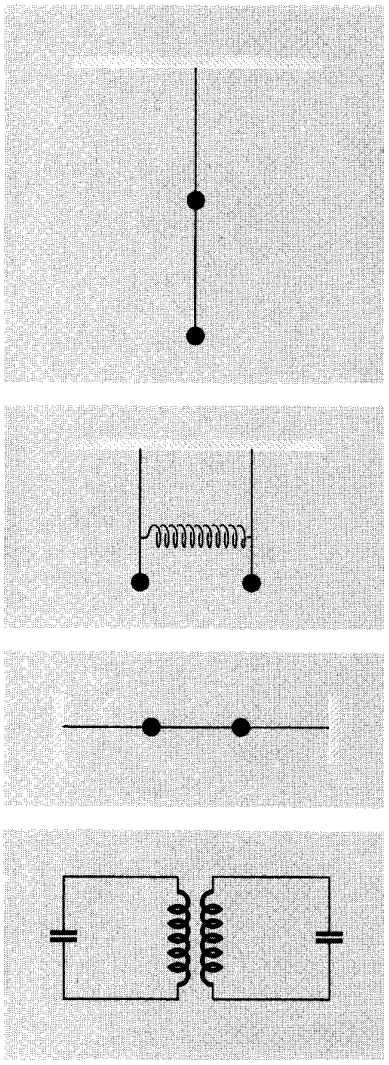


Fig. 2.5: Examples of systems of two degrees of freedom

Modes of Oscillations

- The general motion of a system with two degrees of freedom can have a very complicated appearance; parts may not move with simple harmonic motion. However, we will see that for the system with two degrees of freedom, the most general motion is linear superposition of two independent simple harmonic motions.
- These simple harmonic motions are called *normal modes* or simply *modes*. By suitable initial conditions, we can make system to oscillate in only one of modes. Thus, modes are uncoupled, even though though the moving parts are not.

Properties of Modes

- For motion of any system in a single one of its modes:
  - ◇ (i) **Each moving part oscillates with the same frequency** ω, i.e, experiences the same return force per unit mass per unit displacement.
    Thus, we *cannot* have a single mode with different frequencies, as in $\psi_a(t) = A\cos\omega_1 t$, $\psi_b(t) = B\cos\omega_2 t$.
  - ◇ **Each moving part passes through the equilibrium positions at the same time in sync. Their phase constants must be the**

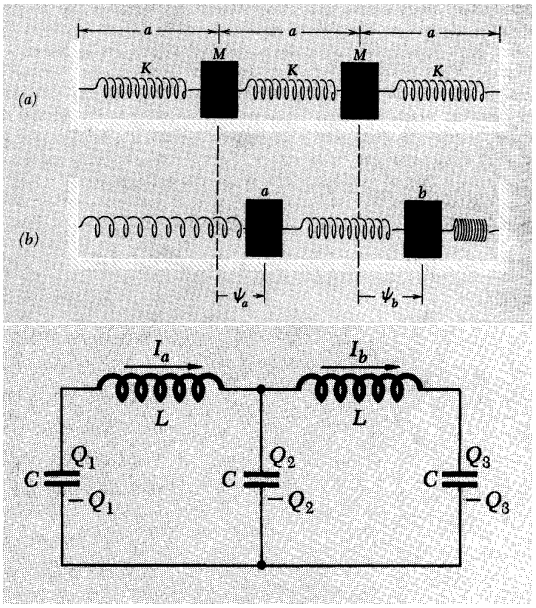


Fig. 2.6: General displacement in systems with two degrees of freedom.

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same apart from a overall sign of the displacements.
Thus, for e.g., we *cannot* have a single mode with different phase constants, as in $\psi_a(t) = A\cos\omega t$, $\psi_b(t) = B\sin\omega t$.

- For example, in a coupled oscillation of two masses A and B, the displacements of the masses must be of the form:

$$\begin{aligned}\psi_a(t) &= A_1\cos(\omega_1 t + \varphi_1),\\ \psi_b(t) &= B_1\cos(\omega_1 t + \varphi_1) = \frac{B_1}{A_1}\psi_a(t),\end{aligned}$$

where the only difference can be in the constants A_1 and B_1. For the sake of convenience, we will allow the ratio B_1/A_1 to have both signs.

- It is easy to see the they pass through the equilibrium positions at the same time. Let $A_1 = A$, $\omega_1 = \omega$, and $\varphi_1 = \varphi$. Then

$$\dot\psi_a = \omega A\sin(\omega t + \varphi), \qquad \dot\psi_b = \omega B\sin(\omega t + \varphi).$$

Both the masses pass through equilibrium at the same time when their velocities become zero at the same time:

$$\dot\psi_a(t) = 0, \qquad \dot\psi_b(t) = 0.$$

This happens whenever

$$\omega t + \varphi = n\pi \qquad \text{or}$$

$$t = \left(\frac{n}{2} - \frac{\varphi}{2\pi}\right)T, \qquad \text{(here, } T \text{ is the period)},$$

regardless of the relative magnitudes or signs of A and B, provided both are non-zero.

- Similarly for mode 2, the motion of two degrees of freedom a and b may be written as

$$\begin{aligned}\psi_a(t) &= A_2\cos(\omega_2 t + \varphi_2),\\ \psi_b(t) &= B_2\cos(\omega_2 t + \varphi_2) = \frac{B_2}{A_2}\psi_a(t).\end{aligned}$$

- The most general solution for the moving parts may be written as

$$\begin{aligned}\psi_a(t) &= A_1\cos(\omega_1 t + \varphi_1) + A_2\cos(\omega_2 t + \varphi_2),\\ \psi_b(t) &= B_1\cos(\omega_1 t + \varphi_1) + B_2\cos(\omega_2 t + \varphi_2).\end{aligned}$$

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2.3.2 Example: Two-dimensional harmonic oscillator

Example 2.4. Two-dimensional harmonic oscillator. Refer to the figure below.

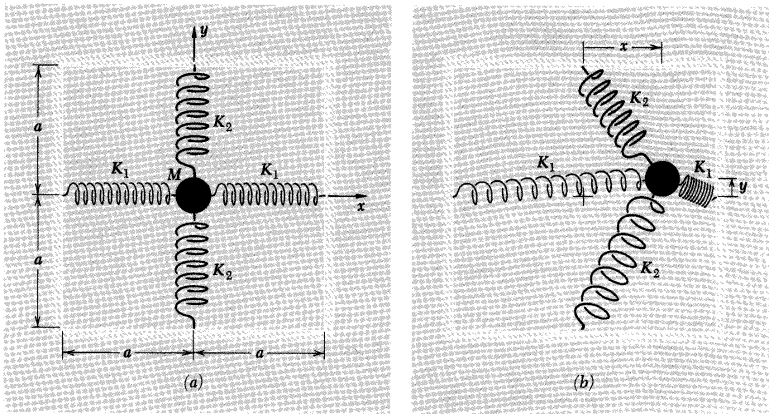


Fig. 2.7: 2D harmonic oscillator illustration

- When the free length a_0 of the spring equals its equilibrium length a, in the small oscillation approximations, as seen earlier, up to first order in the displacements, the forces along x and y directions are

$$F_x = -2K_1 x, \qquad F_y = -2K_2 y.$$

- Equations for x and y motions are

$$M\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -2K_1 x, \qquad \text{and} \qquad M\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -2K_2 y.$$

  In this simple case, both the equations are decoupled and simpler.

- The solutions are simple harmonic motions with frequencies dependent on the spring constants:

$$\begin{aligned} x &= A_1\cos(\omega_1 t + \varphi_1), \qquad \omega_1^2 = \frac{2K_1}{M}, \\ y &= B_2\cos(\omega_2 t + \varphi_2), \qquad \omega_2^2 = \frac{2K_2}{M}. \end{aligned}$$

- We see that here x motion and y motion are not coupled. Each motion is a harmonic oscillation with its own frequency.

- Thus the x motion corresponds to one normal mode of oscillation, the y motion to the other. The x mode has amplitude A_1 and phase constant φ_1 that depend only on the initial values $x(0)$ and $\dot{x}(0)$, i.e., the x displacement and velocity at time $t = 0$. Similarly the y mode has amplitude B_2 and phase constant φ_2 that depend only on the initial values $y(0)$ and $\dot{y}(0)$.

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2.3.3 Systematic method of solution for coupled oscillations

Example 2.5. Two-dimensional harmonic oscillator – at connected at oblique angles – systematic method

- Imagine the two-dimensional oscillator (mass-springs system) we considered in the previous section is rotated by an angle and fixed to the wall as shown in the Fig.(2.8).

- We expect that x and y motions get coupled and result in the general two coupled first-order linear homogeneous equations in the default system coordinates x and y given by

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -a_{11}x - a_{12}y, \tag{2.21}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -a_{11}x - a_{12}y. \tag{2.22}$$

  These equations mean that a displacement along x direction will result in a restoring force in all four springs and will result in a motion also along y axis. Similarly, a displacement along y axis will also result in a motion along x axis. Said concisely, motion along one direction is necessarily correlated to the motion along the other direction. This is meaning of "coupling".

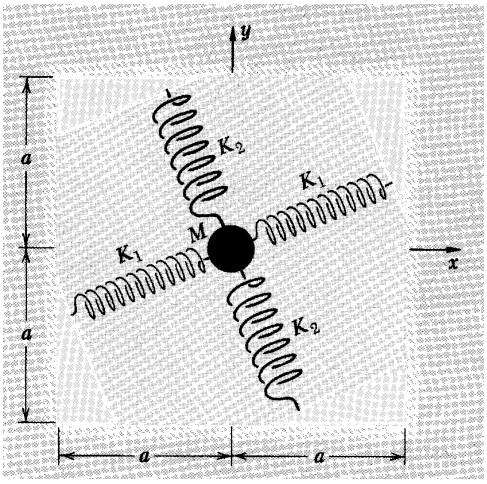


Fig. 2.8: Transverse oscillations. (a) Equilibrium configuration. (b) General configuration (for motion along x).

- Now assume that we have oscillation in a single normal mode – both x and y oscillate with harmonic motion with the same frequency and same phase constant:

$$x = A\cos(\omega t + \varphi), \qquad y = B\cos(\omega t + \varphi). \tag{2.23}$$

  Here, ω and B/A are unknown.

- Then,

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 A\cos(\omega t + \varphi) = -\omega^2 x, \tag{2.24}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\omega^2 B\cos(\omega t + \varphi) = -\omega^2 y. \tag{2.25}$$

  Substituting into the Eqs.(2.21) and (2.21), we get

$$(a_{11} - \omega^2)x + a_{12}y = 0, \tag{2.26}$$

$$a_{21}x + (a_{22} - \omega^2)y = 0. \tag{2.27}$$

- The ratio y/x from both the equations must be the same:

$$\boxed{\frac{y}{x} = \frac{B}{A} = \frac{(\omega^2 - a_{11})}{a_{12}} = \frac{a_{21}}{(\omega^2 - a_{22})}} \quad \Rightarrow \tag{2.28}$$

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$$\boxed{(\omega^2 - a_{11})(\omega^2 - a_{22}) - a_{12}a_{21} = 0.} \tag{2.29}$$

- The same equation can also be obtained by setting the determinant of the coefficient matrix of the above linear equations to 0:

$$\begin{vmatrix} a_{11} - \omega^2 & a_{12} \\ a_{21} & a_{22} - \omega^2 \end{vmatrix} = 0. \tag{2.30}$$

- The resultant quadratic equation in ω^2 gives two solutions, say, ω_1^2 and ω_2^2 from which we obtain the normal mode frequencies ω_1 and ω_2.

- These two normal mode frequencies give solutions. By substituting these frequencies into the equation 2.28, we get the amplitude ratios for each mode:

$$\left(\frac{y}{x}\right)_{\text{mode 1}} = \left(\frac{B}{A}\right)_{\text{mode 1}} = \frac{B_1}{A_1} = \frac{(\omega_1^2 - a_{11})}{a_{12}}, \tag{2.31}$$

$$\left(\frac{y}{x}\right)_{\text{mode 2}} = \left(\frac{B}{A}\right)_{\text{mode 2}} = \frac{B_2}{A_2} = \frac{(\omega_2^2 - a_{11})}{a_{12}}. \tag{2.32}$$

- The most general solution for each mass is the linear superposition of the oscillations from each mode:

$$\begin{aligned} x(t) &= x_1(t) + x_2(t) \\ &= A_1\cos(\omega_1 t + \varphi_1) + A_2\cos(\omega_2 t + \varphi_2), \end{aligned} \tag{2.33}$$

$$\begin{aligned} y(t) &= y_1(t) + y_2(t) \\ &= B_1\cos(\omega_1 t + \varphi_1) + B_2\cos(\omega_2 t + \varphi_2). \end{aligned} \tag{2.34}$$

- From the initial conditions $x(0)$, $\dot{x}(0)$, $y(0)$, and $\dot{y}(0)$ we can fix four undetermined coefficients A_1, A_2, φ_1, and φ_2. The coefficients B_1 and B_2 are determined using (2.31) and (2.32).

✓ **Exercise 2.3.** A numerical example of coupled oscillations.

(a) Determine the normal mode frequencies for a system of coupled oscillations described the system linear ODEs given by

$$\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -3\psi_a - \psi_b; \qquad \frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -\psi_a - 3\psi_b.$$

Steps:

(i) Start with trial solutions of the form

$$\psi_a = A\cos(\omega t + \varphi), \qquad \psi_a = B\cos(\omega t + \varphi).$$

Both the masses have same frequencies and the same phases, apart from a overall sign; they pass through the equilib-

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rium positions at the same time. Substitute them into the linear 2nd order ODEs, and obtain a set of linear equations for ψ_a and ψ_b.

(ii) Then obtain two equations for the ratio of the displacements: ψ_b/ψ_a.

(iii) Both the equations must be equivalent. So, setting them equal gives a quadratic equation for ω^2. Arrive at this quadratic equation.

(iv) Solve the quadratic equation for ω^2 and obtain two normal mode frequencies.

(v) For each mode frequency, evaluate the ratio B/A. Then express, B in terms of A for each mode.

(vi) In the last step, write down the displacements of the masses for each mode. Finally write down the general displacements ψ_a and ψ_b of masses, similar to the Eqs. (2.33) and (2.34).

(b) If the initial conditions be given by

$$\psi_a(0) = 2, \quad \dot{\psi}_a(0) = 0; \qquad \psi_b(0) = 0, \quad \dot{\psi}_b(0) = 0.$$

find A_1, B_1, φ_1 and φ_2, and write down the most general solution.

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2.4 Oscillations of two coupled masses

2.4.1 Longitudinal oscillations of two coupled masses

Example 2.6. The two masses M slide on a frictionless table. The three springs are massless and identical, each with spring constant K. Equilibrium and a general displacement configuration are shown in the figure below.

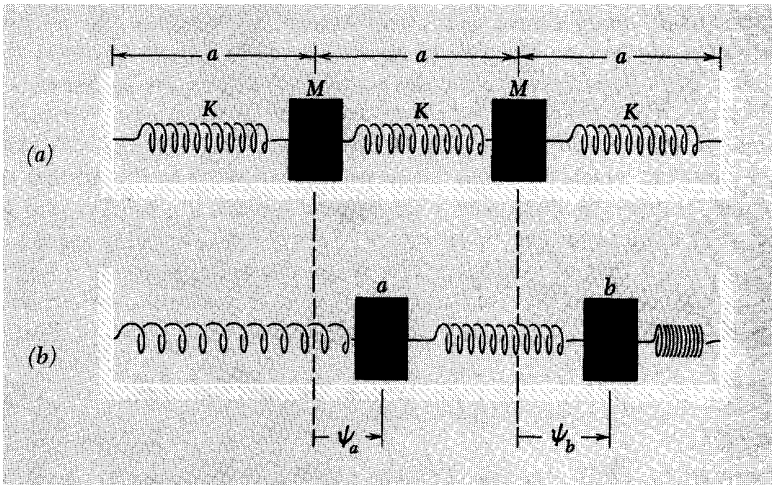


Fig. 2.9: Longitudinal oscillations. (a) Equilibrium. (b) General configuration.

- **Kinematics and Dynamics**:
  - ◇ Equilibrium positions of the blocks: $z_{a0} = a,\ z_{b0} = 2a$.
  - ◇ Displacements of the blocks from their equilibrium positions:

$$\psi_a,\ \psi_b.$$

  - ◇ Position of the block A when displaced from equilibrium positions:

$$z_a = a + \psi_a, \qquad z_b = 2a + \psi_b.$$

  - ◇ Accelerations of the masses. Since a and $2a$ are constant additive terms in z_a and z_b,

$$\text{acc}_a = \frac{\mathrm{d}^2 z_a}{\mathrm{d}t^2} = \frac{\mathrm{d}^2 \psi_a}{\mathrm{d}t^2}, \qquad \text{acc}_b = \frac{\mathrm{d}^2 z_b}{\mathrm{d}t^2} = \frac{\mathrm{d}^2 \psi_b}{\mathrm{d}t^2}.$$

  - ◇ Change in the length and tension in the spring 1:

$$\Delta l_1 = z_a - a = \psi_a, \qquad T_1 = K\psi_a$$

  - ◇ Change in the length and tension in the spring 2:

$$\Delta l_2 = z_b - z_a - a = \psi_b - \psi_a, \qquad T_2 = K(\psi_b - \psi_a)$$

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- ◇ Change in the length and tension in the spring 3 (when $\psi_a < 0$, its compression and the restoring force in the direction of restoring the length of the spring):

$$\Delta l_3 = -\psi_b, \qquad T_3 = -K\psi_b$$

- ◇ Net restoring force on mass A:

$$F_a = T_2 - T_1 = K(\psi_b - \psi_a) - K\psi_a = -2K\psi_a + K\psi_b.$$

- ◇ Net restoring force on mass B:

$$F_b = T_3 - T_2 = -K\psi_b - K(\psi_b - \psi_a) = K\psi_a - 2K\psi_b.$$

- Equations of Motion.
  - ◇ EoM for Mass A:

$$M\,\mathrm{acc}_a = M\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -K\psi_a + K(\psi_b - \psi_a). \tag{2.35}$$

  - ◇ EoM for Mass B: Similarly,

$$M\,\mathrm{acc}_b = M\frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2} = -K\psi_b - K(\psi_b - \psi_a). \tag{2.36}$$

- *Identification of modes by guess work.* When both the masses move in sync with the same frequency the combined motion corresponds to a mode of coupled oscillation. Then, both the masses must experience the same restoring force per unit displacement per unit mass.

- **Mode 1**:
  - ◇ Both masses move in sync such that their displacements are equal

$$\psi_a = \psi_b.$$

  - ◇ Then, the coupling disappears and the EoMs reduce to a single equation

$$\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -\frac{K}{M}\psi_a, \quad = \quad -\frac{K}{M}\psi_b = \frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2}.$$

  - ◇ Dividing by masses and displacements, we get

$$\frac{1}{\psi_a}\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -\frac{K}{M} = \frac{1}{\psi_b}\frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2} = \text{the same restoring force per unit displacement per unit mass}$$

  - ◇ Therefore, both the masses oscillate with the *mode 1 frequency*

$$\omega_1^2 = \frac{K}{M}.$$

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- **Mode 2**:
  - ◇ Both masses move in sync such that their displacements are equal and opposite:

$$\psi_a = -\psi_b.$$

  - ◇ Then, the coupling disappears and the EoMs reduce to the single equation

$$\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -\frac{3K}{M}\psi_a \quad = \quad \frac{3K}{M}\psi_b = -\frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2}.$$

  - ◇ Therefore, both the masses oscillate with the *mode 2 frequency*

$$\omega_2^2 = \frac{3K}{M}.$$

  Both the motions are illustrated in the figure below.

Exercise 2.4. (a) **Using the systematic method**, determine the mode frequencies and amplitude ratios of mass in each mode for the longitudinal oscillations two coupled masses, as discussed in Example 2.6 above. (b) Perform the numerical calculations, taking, $K = 0.45\,\mathrm{N/m}$, $M = 50\,\mathrm{g}$.

Hints/Algorithm:

i) Write down the two coupled 2nd order linear ODE for ψ_1 and ψ_2;

ii) deduce/express/identify the coefficients $a_{11}, \ldots, a_{22}$ in terms of the parameters of the coupled oscillators problem.

iii) find the characteristic equation for ω^2;

iv) solve the quadratic equation to find normal mode frequencies ω_1^2 and ω_2^2;

v) find the ratio of amplitudes (B/A) for each mode by substituting the values of ω^2 for each mode, and see if you get back the same results as in the guesswork method for both frequencies and ratio of displacements;

vi) Finally calculate the numerical values for the given numerical mass-spring parameters.

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2.4.2 Transverse oscillations of two coupled masses

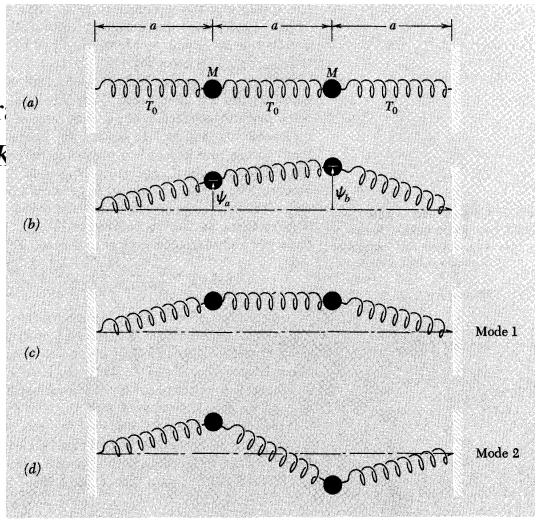


Fig. 2.10: Transverse oscillations. (a) Equilibrium. (b) General configuration. (c) Mode with lower frequency. (d) Mode with higher frequency.

Example 2.7. The two masses M slide on a frictionless table in the tr
The three springs are massless and identical, each with spring constant K
a general displacement configuration are shown in the figure below.

- **Kinematics and Dynamics**:
  - ◇ Equilibrium positions of the blocks along the z axis:

$$z_{a0} = a,\ z_{b0} = 2a.$$

  - ◇ Displacements of the blocks from their equilibrium positions in the transverse direction:

$$\psi_a,\ \psi_b$$

  - ◇ Accelerations of the masses.

$$\text{acc}_a = \frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2}, \qquad \text{acc}_b = \frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2}.$$

  - ◇ Angles springs make with z (horizontal) axis:

$$\theta_1 \approx \frac{\psi_a}{a}; \qquad \theta_2 = \frac{\psi_b - \psi_a}{a}; \qquad \theta_3 \approx \frac{\psi_b}{a}$$

  - ◇ change in the lengths and tension in the springs in the small oscillation approximation:

$$\Delta l_1 \approx \Delta l_2 \approx \Delta l_3 \approx a - a_0, \qquad T_1 \approx T_2 \approx T_3 \approx K(a - a_0) = T_0.$$

  - ◇ Net force on masses:

$$\begin{aligned} F_a &= T_2 \sin\theta_2 - T_1 \sin\theta_1 \approx T_0\theta_2 - T_0\theta_1 \approx -T_0\frac{\psi_a}{a} + T_0\frac{\psi_b - \psi_a}{a} \\ F_b &= -T_2 \sin\theta_2 - T_3 \sin\theta_3 \approx -T_0\theta_2 - T_0\theta_3 \approx -T_0\frac{\psi_b}{a} - T_0\frac{\psi_b - \psi_a}{a} \end{aligned}$$

- Equations of Motion
  - ◇ EoM for Masses A and B are

$$\begin{aligned} M\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} &= -\frac{T_0}{a}\psi_a + \frac{T_0}{a}(\psi_b - \psi_a), \\ M\frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2} &= -\frac{T_0}{a}\psi_b - \frac{T_0}{a}(\psi_b - \psi_a). \end{aligned}$$

  - ◇ Above two equations look the same as the Eqs. (2.35) and (2.36), except for $K \to T_0/a$.
  - ◇ They have similar mode frequencies and the same ratio of amplitudes of oscillations of masses.

- **Mode 1**:

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- ◇ Please refer to the textbook. Both masses move in sync such that their displacements are equal

$$\psi_a = \psi_b.$$

and both the masses oscillate with the *mode 1 frequency*

$$\omega_1^2 = \frac{K}{M}.$$

- **Mode 2**:
  - ◇ Both masses move in sync such that their displacements are equal and opposite:

$$\psi_a = -\psi_b.$$

and they oscillate with the *mode 2 frequency*

$$\omega_2^2 = \frac{3K}{M}.$$

Both the motions are illustrated in the figure above.

- **Exercise 2.5.** (a) Using the systematic method, determine the mode frequencies and amplitude ratios of mass in each mode for the transverse oscillations two coupled masses, as discussed in Example 2.7 above. (b) Perform the numerical calculations, taking, $a = 10\,\text{cm}$, $T_0 = 4.5 \times 10^{-3}\,\text{N}$, $M = 50\,\text{g}$.

Hints/Algorithm:

i) Write down the two coupled 2nd order linear ODE for ψ_1 and ψ_2;

ii) deduce/express/identify the coefficients $a_{11}, \ldots, a_{22}$ in terms of the parameters of the coupled oscillators problem.

iii) find the characteristic equation for ω^2;

iv) solve the quadratic equation to find normal mode frequencies ω_1^2 and ω_2^2;

v) find the ratio of amplitudes (B/A) for each mode by substituting the values of ω^2 for each mode, and see if you get back the same results as in the guesswork method for both frequencies and ratio of displacements;

vi) Finally calculate the numerical values for the given numerical mass-spring parameters.

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2.4.3 Coupled Pendulums

Example 2.8. Coupled Pendulum – an elementary analysis of the problem

- Please refer to Fig.2.11 on the right.
- Referring to the longitudinal coupled mass oscillations, and the simple pendulum, we can arrive the equations of motions

$$\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -\frac{g}{l}\psi_a + \frac{K}{M}(\psi_b - \psi_a), \tag{2.37a}$$

$$\frac{\mathrm{d}^2\psi_b}{\mathrm{d}t^2} = -\frac{g}{l}\psi_b - \frac{K}{M}(\psi_b - \psi_a). \tag{2.37b}$$

- Mode 1 oscillations – simpler analysis :
  - Consider the motion in which the displacements of both the masses are identical:

$$\psi_a = \psi_b.$$

    In this pattern of movement, the length of the middle spring remains constant and both the masses move together as if it is a single rigid unit of mass $2M$. The restoring force arises only from gravity.
  - By putting $\psi_b = \psi_a$ in Eq.(2.37a),

$$\frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} = -\frac{g}{l}\psi_a\psi_a \qquad \Rightarrow \qquad \omega_1^2 = \frac{g}{l},$$

    we see that ODE for ψ_a becomes equivalent to that of a decoupled independent oscillator. The equation of motion, restoring force and frequency all become the same for both the masses.
  - The general solution for motion is

$$\psi_1 = A_1 \cos(\omega_1 t + \varphi_1).$$

- Mode 2 oscillations – simpler analysis:
  - In a similar manner, now consider the motion in which the displacements of both the masses are equal and opposite,

$$\psi_b = -\psi_a.$$

    In this pattern of movement, the the spring gets compressed or stretched, and restoring force comes from both spring and gravity.

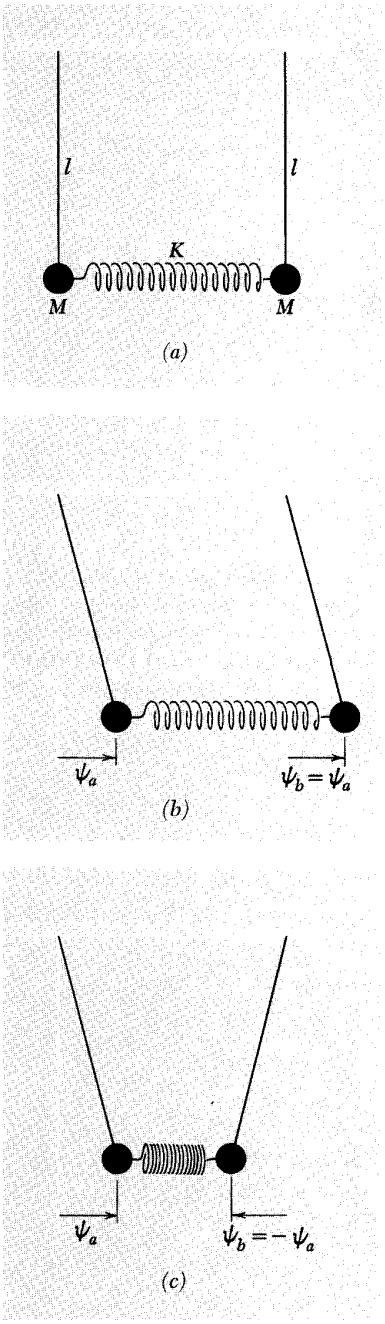


Fig. 2.11: Coupled identical pendulums. (a) Equilibrium configuration. (b) Mode with lower frequency. (c) Mode with higher frequency.

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- ◇ By putting $\psi_b = -\psi_a$ in Eq.(2.37a), we get

$$\begin{aligned} \frac{\mathrm{d}^2\psi_a}{\mathrm{d}t^2} &= -\left(\frac{g}{l}+\frac{2K}{M}\right)\psi_a \qquad \Rightarrow \\ \omega_2^2 &= \frac{g}{l}+\frac{2K}{M}, \end{aligned}$$

  we see once again that the ODE for ψ_a becomes equivalent to that of a decoupled independent oscillator – the equation of motion, restoring force and frequency all become the same for both the masses.

- ◇ The general solution for motion is

$$\psi_2 = A_2\cos(\omega_2 t+\varphi_2).$$

- The general solution for ψ_a and ψ_b, displacements of masses.
  - ◇ For mass A: The most general solution is simply a linear superposition of displacements from both the modes the free parameters A_1, φ_1 and A_2, φ_2:

$$\begin{aligned} \psi_a &= \left(\psi_a\right)_{\text{mode 1}} + \left(\psi_a\right)_{\text{mode 2}} \\ & A_1\cos(\omega_1 t+\varphi_1) + A_2\cos(\omega_2 t+\varphi_2), \end{aligned} \tag{2.38}$$

  - ◇ For mass B, since the displacement ratios for the two modes are given by

$$\left(\frac{\psi_b}{\psi_a}\right)_{\text{mode1}} = 1, \qquad \left(\frac{\psi_b}{\psi_a}\right)_{\text{mode2}} = -1,$$

    we get,

$$\psi_b = A_1\cos(\omega_1 t+\varphi_1) - A_2\cos(\omega_2 t+\varphi_2). \tag{2.39}$$

  - ◇ From the above equation, we obtain the relationship between the pure mode displacements and those of the masses:

$$\psi_1 = \tfrac{1}{2}(\psi_a+\psi_b), \tag{2.40a}$$

$$\psi_2 = \tfrac{1}{2}(\psi_a-\psi_b). \tag{2.40b}$$

- Solving for the undetermined coefficients.
  - ◇ Initial displacements and velocities of the masses will be given:

$$\psi_a(0), \quad \psi_b(0), \qquad \dot{\psi}_a(0), \quad \dot{\psi}_b(0).$$

  - ◇ From these conditions, we get the conditions for the pure mode

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solutions:

$$\psi_1(0) = \tfrac{1}{2}[\psi_a(0)+\psi_b(0)], \tag{2.41a}$$
$$\dot{\psi}_1(0) = \tfrac{1}{2}[\dot{\psi}_a(0)+\dot{\psi}_b(0)] \tag{2.41b}$$
$$\psi_2(0) = \tfrac{1}{2}[\psi_a(0)-\psi_b(0)], \tag{2.41c}$$
$$\dot{\psi}_2(0) = \tfrac{1}{2}[\dot{\psi}_a(0)-\dot{\psi}_b(0)] \tag{2.41d}$$

- From the conditions for ψ_1, we can solve for A_1, φ_1 and from those for ψ_2 we can solve for A_2, φ_2.

Exercise 2.6. Normal modes of coupled pendulums.

(a) For $g = 9.8\,\mathrm{m/s^2}$,$l = 1\,\mathrm{m}$, $K = 0.25\,\mathrm{N/m}$, $M = 250\,\mathrm{g}$, find the normal mode frequencies ω_1 and ω_2 of the coupled pendulums.

(b) If the mass A is initially displaced by 3 cm to the right, as indicated in the figure, find the displacements of the both masses at time t. This means that the initial displacement and velocities of the masses are given by

$$\psi_a(0) = 4\,\mathrm{cm}, \qquad \dot{\psi}_a(0) = 0\,\mathrm{cm/s},$$
$$\psi_b(0) = 0\,\mathrm{cm}, \qquad \dot{\psi}_b(0) = 0\,\mathrm{cm/s}.$$

Using this, calculate the initial conditions on ψ_1, $\dot{\psi}_1$, ψ_2 and $\dot{\psi}_2$ in Eqs. (2.41a) and (2.41c), and hence determine A_1, φ_1 and A_2, φ_2. Finally find $\psi_a(t)$ and $\psi_b(t)$.
Start by letting $\psi_a(0) = 2A$. Then show that

$$A_1 = A_2 = A, \qquad \varphi_1 = \varphi_2 = 0,$$

and

$$\psi_a(t) = A\cos\omega_1 t + A\cos\omega_2 t, \qquad \psi_b(t) = A\cos\omega_1 t - A\cos\omega_2 t.$$

Finally substitute numerical values.

Exercise 2.7. Apply the systematic method to arrive at the above mode frequencies and ratios of displacements. Use similar hints given earlier.

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2.5 Beats

- When coupled oscillations of two masses are involved, the displacement of a mass is a superposition (the linear combination) of two normal mode oscillations which are simple harmonic oscillations:

$$\psi = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2).$$

  - ◇ In this section, we will see what happens when the two normal mode frequencies ω_1 and ω_2 of such coupled oscillations. This will in turn allows us a means to study and characterize a concrete example of coupled oscillations and make measurements them.
  - ◇ Such a superposition of oscillations occur in experiments in physics related to sound.

- When a mechanical moving part is a superposition of two harmonic oscillations having different angular frequencies ω_1 and ω_2, say corresponding to two normal modes of a coupled system, like two tuning forks, resulting sound wave alternates between loud and soft sound periodically. This periodic variation in loudness is called ***beats***. Such motion is illustrated in Fig. 2.12 below.

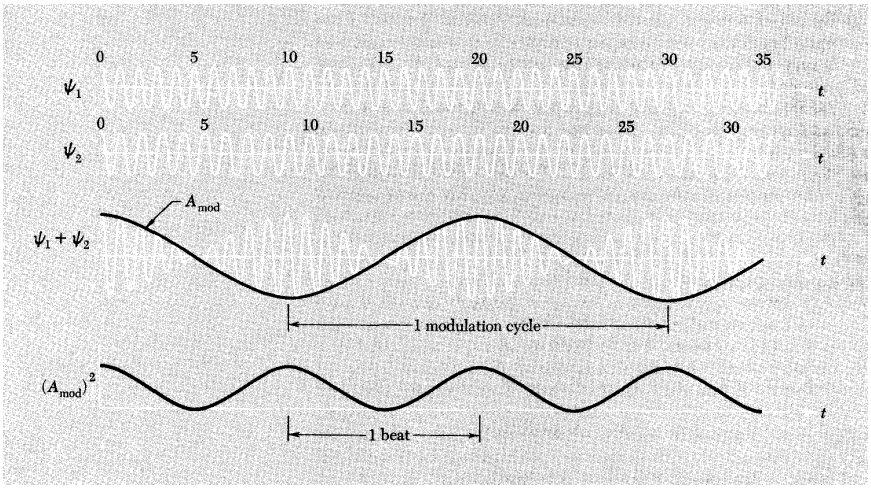


Fig. 2.12: Beats. ψ_1 and ψ_2 are the pressure variations at your ear produced by two tuning forks with frequency ratio $\nu_1/\nu_2 = 10/9$. The total pressure is the superposition , $\psi_1 + \psi_2$, which is an "almost harmonic" oscillation at frequency ν_{av} with slowly varying amplitude $A_{mod}(t)$. The loudness is proportional to $(A_{mod})^2$ and consists of a constant (average value) plus a sinusoidal variation at the beat frequency. The beat frequency is twice the modulation frequency.

- Proof: It is simple – uses basic trigonometry.
  - ◇ To make calculations simpler, let us assume that the two harmonic oscillations, having angular frequencies ω_1 and ω_2, have

DRAFT

DRAFT

the *same amplitude* and have the *same phase constant*, which may be taken to be zero:

$$\psi_1 = A\cos\omega_1 t, \qquad \psi_2 = A\cos\omega_2 t.$$

◇ Let's consider the simplest superposition of the two oscillations – sum of the two oscillations:

$$\psi = \psi_1 + \psi_2 = A\cos\omega_1 t + A\cos\omega_2 t. \tag{2.42}$$

This happens, say, when the mass A is pulled to the right with the initial displacement of $2A$, i.e, $\psi_a(0) = 2A$, while all the other initial conditions are to 0. See the exercise (2.6).

◇ ***Average frequency and modulation frequency***: We will define them as we shall soon encounter them. The average and modulation frequencies are defined by

$$\omega_{\rm av} = \tfrac{1}{2}(\omega_1 + \omega_2), \qquad \omega_{\rm mod} = \tfrac{1}{2}(\omega_2 - \omega_1). \tag{2.43}$$

◇ The mode frequencies in turn are easily seen to be related to the average and modulation frequencies, $\omega_{\rm av}$ and $\omega_{\rm mod}$, by

$$\omega_1 = \omega_{\rm av} - \omega_{\rm mod}, \qquad \omega_2 = \omega_{\rm av} + \omega_{\rm mod}. \tag{2.44}$$

◇ Then, using the cosine addition formula, the superposition simplifies to

$$\begin{aligned} \psi &= A\cos\omega_1 t + A\cos\omega_2 t \\ &= A\cos(\omega_{\rm av}t - \omega_{\rm mod}t) + A\cos(\omega_{\rm av}t + \omega_{\rm mod}t) \\ &= \underbrace{[2A\cos\omega_{\rm mod}t]}_{\text{modulated amplitude}\,A_{\rm mod}(t)} \cos\omega_{\rm av}t. \end{aligned} \tag{2.45}$$

◇ Thus, the superposition two waves may be expressed in the form:

$$\psi = A_{\rm mod}(t)\cos\omega_{\rm av}t, \tag{2.46}$$

where

$$A_{\rm mod}(t) = 2A\cos\omega_{\rm mod}t. \tag{2.47}$$

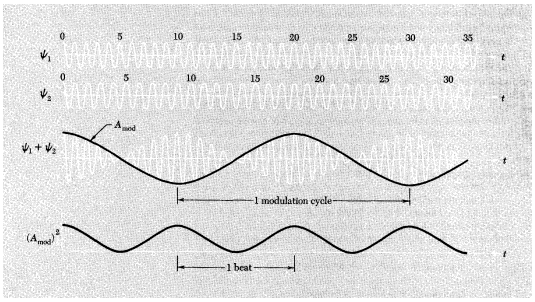
The Eqs.(2.46) and (2.47) may be regarded as representing an oscillation at the mean angular frequency $\omega_{\rm av}$, with an amplitude $A_{\rm mod}$ that is not constant but rather varies with time according to Eq.(2.47).

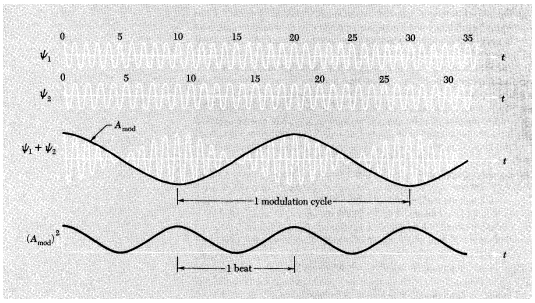
◇ Even though the expression in Eq. (2.46) is exactly equivalent to (2.42), it is most useful to write the superposition in the form of Eq.(2.42) when ω_1 and ω_2 are of comparable magnitude. The variation in the displacement defined by (2.46), together with

DRAFT

DRAFT

(2.47), is illustrated again in the figure on the right.





- ***Square of the amplitude beats frequency***.
  - ◇ Human ears can only detect variations in intensity of sound waves, and not the amplitude. Since intensities are proportional to square of the amplitude, our ears essentially detect only the square of the modulate amplitude of the superposed oscillations that reach our ears:

$$\begin{aligned}[A_{\rm mod}(t)]^2 &= 4A^2\cos^2(\omega_{\rm mod}t)^2,\\ &= 2A^2[1+\cos(2\omega_{\rm mod}t)],\\ &= 2A^2[1+\cos(\omega_{\rm beat}t)].\end{aligned}$$

  - ◇ Thus, the frequency of beats of the superposed oscillations are

$$\boxed{\omega_{\rm beat} = 2\omega_{\rm mod} = |\omega_2 - \omega_1|.}$$

  - ◇ Variations of superposed oscillations and square of modulated amplitude are illustrated in Fig. 2.12.

Exercise 2.8. Derive the following expression for difference in the displacements:

$$\psi = \psi_1 - \psi_2 = A\cos\omega_1 t - A\cos\omega_2 t = B_{\rm mod}(t)\sin\omega_{\rm av}t,$$

where now,

$$B_{\rm mod}(t) = 2A\sin\omega_{\rm mod}t, \tag{2.48}$$

with $\omega_{\rm mod}$ and $\omega_{\rm av}$ are the same as in before, given in Eqs. (2.44).

DRAFT

DRAFT

Example 2.9. Energy oscillations of the masses in a pair of coupled pendulums.

- In a coupled pendulum problem, let the initial conditions be

$$\begin{aligned} \psi_a(0) &= 2A\,, \qquad \dot{\psi}_a(0) = 0\,, \\ \psi_b(0) &= 0\ \ , \qquad \dot{\psi}_b(0) = 0\,. \end{aligned}$$

  Find the energies of masses $E_a(t)$ and $E_b(t)$ in the approximation that modulated amplitudes vary very slowly compared to the oscillations at the average frequency.

- **(i) Solving the undetermined constants**
  - ◇ The normal mode oscillations may be expressed in the form (please see Eqs. (2.40a) and (2.40b))

$$\psi_1 = \frac{1}{2}(\psi_a + \psi_b) = A_1 \cos(\omega_1 t + \varphi_1), \tag{2.49}$$

$$\psi_2 = \frac{1}{2}(\psi_a - \psi_b) = A_2 \cos(\omega_1 t + \varphi_1). \tag{2.50}$$

  - ◇ Initial conditions on ψ_1 and ψ_2 then become:

$$\begin{aligned} \psi_1(0) &= A, \qquad \dot{\psi}_1(0) = 0, \\ \psi_2(0) &= A, \qquad \dot{\psi}_2(0) = 0. \end{aligned}$$

  - ◇ It was shown in the class that these initial conditions imply

$$A_1 = A_2 = A, \qquad \varphi_1 = \varphi_2 = 0.$$

- **(ii) Displacements of masses**
  - ◇ The two Eqs. (2.49) and (2.50) imply,

$$\psi_a = \psi_1 + \psi_2 = A\cos\omega_1 t + A\cos\omega_2 t, \tag{2.51}$$

$$\psi_b = \psi_1 - \psi_2 = A\cos\omega_1 t - A\cos\omega_2 t. \tag{2.52}$$

  - ◇ From the expression for ψ_a in Eq.(2.51), using the same calculations leading to Eq. (2.45), the displacement of mass A becomes

$$\psi_a = \psi_1 + \psi_2 = A[\cos\omega_1 t + \cos\omega_2 t] = A_{\rm mod}(t)\cos\omega_{\rm av} t,$$

    where $A_{\rm mod}$, $\omega_{\rm mod}$ and $\omega_{\rm av}$ have already been defined.
  - ◇ Similarly, displacement for for mass B given in (2.52), using result in Eq. (2.48), from exercise (2.8),becomes

$$\psi_b = \psi_1 - \psi_2 = A[\cos\omega_1 t - \cos\omega_2 t] = B_{\rm mod}(t)\sin\omega_{\rm av} t,$$

    where now, $B_{\rm mod}(t) = 2A\sin\omega_{\rm mod} t$.

- **(iii) Energies of the masses**. Energy of each mass can be approx-

DRAFT

DRAFT

imated to the **twice** the kinetic energy associated with fast oscillations, in analogy with results for a single pendulums.

- Ignoring the much smaller contribution from the slowly varying of modulation amplitude $A_{\rm mod}(t)$,

$$\begin{aligned} E_a &\approx 2\times\frac{1}{2}M\omega_{\rm av}^2\langle A_{\rm mod}^2(t)\sin^2\omega_{\rm av}t\rangle \qquad \text{(ignoring the term with } \dot{A}_{\rm mod}^2 \propto \omega_{\rm mod}^2 \ll \omega_{\rm av}^2\text{)} \\ &\approx M\omega_{\rm av}^2 A_{\rm mod}^2(t)\underbrace{\langle\cos^2\omega_{\rm av}t\rangle}_{1/2} \qquad \text{(we assumed } A_{\rm mod}(t) \text{ varies slowly over } T_{\rm av}\text{)} \\ &= \frac{1}{2}M\omega_{\rm av}^2 A_{\rm mod}^2(t) = 2M\omega_{\rm av}^2 A^2\cos^2\omega_{\rm mod}t. \end{aligned}$$

- Similarly, for mass B,

$$E_b = \frac{1}{2}M\omega_{\rm av}^2 B_{\rm mod}^2 = 2M\omega_{\rm av}^2 A^2\sin^2\omega_{\rm mod}t$$

See Fig.2.13 for an illustration.

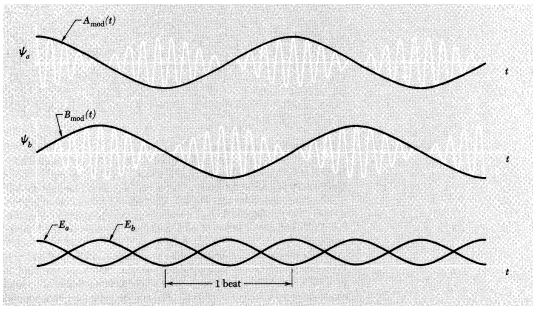


Fig. 2.13: Energy transfer between two weakly coupled identical pendulums. Energy flows back and forth from a to b at the frequency $|\nu_2-\nu_1|$, the beat frequency of the two modes.

- Their total is a constant, as expected:

$$\begin{aligned} E = E_a + E_b &= 2M\omega_{\rm av}^2 A^2[\cos^2\omega_{\rm mod}t + \sin^2\omega_{\rm mod}t \\ &= 2M\omega_{\rm av}^2 A^2 = \text{constant}. \end{aligned}$$

- Their energy difference

$$\begin{aligned} E_a - E_b &= E[\cos^2\omega_{\rm mod}t - A^2\sin^2\omega_{\rm mod}t] \\ &= E\cos 2\omega_{\rm mod}t = E\cos\omega_{\rm beat}t \\ &= E\cos(\omega_2-\omega_1)t. \end{aligned}$$

- Thus, energies of masses oscillate with the beat frequency as given by

$$E_a = \tfrac{1}{2}E[1+\cos(\omega_2-\omega_1)t], \qquad (2.53a)$$
$$E_b = \tfrac{1}{2}E[1-\cos(\omega_2-\omega_1)t]. \qquad (2.53b)$$

- The energy of each mass varies with the beat frequency $2\pi\omega_{\rm beat} = |\nu_2-\nu_1|$ between 0 to E back and forth with an amplitude of half the total energy E. This illustrated in Fig. 2.13.

Exercise 2.9. Numerical example for Beats.

(a) For the frequencies of the waves to be superposed, refer to the exercise 2.6. Find the average, modulation and beats frequencies $\omega_{\rm av}$, $\omega_{\rm mod}$ and $\omega_{\rm beats}$.

(b) When two oscillations are superposed,

(i) What the frequency at which the superposed displacement oscillates in a short time scale?

DRAFT

DRAFT

(ii) What the frequency at which the amplitude of the superposed oscillation modulates?

(iii) What is the frequency at which the energy transfer between the masses oscillates?

Problem 2.1. Systems with more than two degrees of freedom. (Problem 1.17 in Berkeley Physics Vol. 3 Waves). Be aware of the important result contained in this problem.

(a) Write down the three equations for a system of three degrees of freedom analogous to the general equations (2.21) and (2.22). Show that if one assumes a mode, one gets a determinantal equation analogous to Eq. (2.30), except that it is a 3×3 determinant. Show that this gives a cubic equation in the variable ω^2. Since a cubic has three solutions, there are three modes.

(b) Generalize to N degrees of freedom. This constitutes a proof that N modes exist for a system of N degrees of freedom. They must exist, because here you have a prescription for finding them.

DRAFT